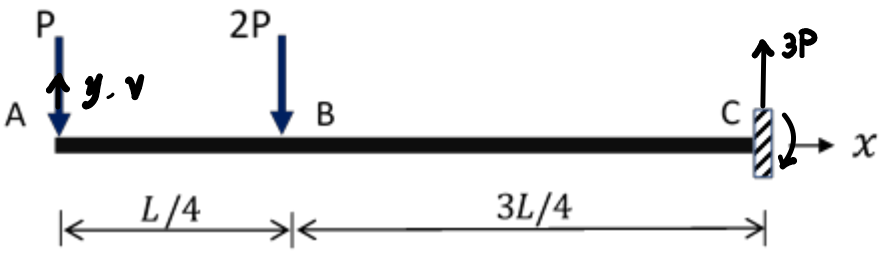
Tutorial Sheet 5B: Elastic Curves

1. Determine the elastic curve for the beam shown below.



$\frac{1}{EI}[-Px - 2P\langle x-\frac{L}{4}\rangle]$

$$\therefore \frac{d^2v}{dx^2} = -\frac{1}{R} = -\frac{M}{EI} = -\frac{(-Px - 2P\langle x-\frac{L}{4}\rangle)}{EI}$$

$$\therefore \frac{dv}{dx} = \int \frac{(-Px - 2P\langle x-\frac{L}{4}\rangle)}{EI}dx = \frac{P}{EI}(-\frac{1}{2}x^2 - \langle x-\frac{L}{4}\rangle^2 + C_1)$$

$$\therefore v = \int \frac{P}{EI}(-\frac{1}{2}\cancel{P}x^2 - \cancel{P}\langle x-\frac{L}{4}\rangle^2 + C_1)dx$$

$$= \frac{P}{EI}(-\frac{1}{6}\cancel{P}x^3 - \frac{1}{3}\cancel{P}\langle x-\frac{L}{4}\rangle^3 + C_1x + C_2)$$

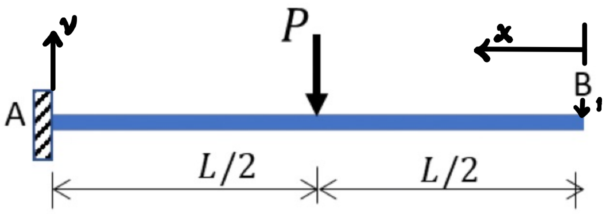
When $x = L$, $\frac{dv}{dx} = 0$ and $v = 0$

$$\therefore C_1 = -\frac{17}{16}L^2$$

$$\therefore C_2 = -\frac{\overset{145}{\cancel{113}}}{192}L^3$$

$$\therefore v(\uparrow) = \frac{P}{EI}(\frac{1}{3}\cancel{P}x^3 + \frac{1}{3}\cancel{P}\langle x-\frac{L}{4}\rangle^3 + \frac{17}{16}L^2x - \frac{\overset{145}{\cancel{113}}}{192}L^3)$$

2. Calculate the maximum deflection using both double integration and unit load approaches. Sketch the deflected shape.



a) $M = -P\langle x - \frac{L}{2}\rangle$

① $\frac{dv^2}{dx^2} = -\frac{1}{R} = -\frac{M}{EI} = \frac{P}{EI}\langle x - \frac{L}{2}\rangle$

$\therefore \frac{dv}{dx} = \int \frac{P}{EI}\langle x - \frac{L}{2}\rangle dx = \frac{P}{EI}(\frac{1}{2}\langle x - \frac{L}{2}\rangle^2 + C_1)$

$\therefore v = \frac{P}{EI}\int(\frac{1}{2}\langle x - \frac{L}{2}\rangle^2 + C_1)dx = \frac{P}{EI}(\frac{1}{6}\langle x - \frac{L}{2}\rangle^3 + C_1 x + C_2)$

When $x = 0$, $v = 0$ and $\frac{dv}{dx} = 0$

$\therefore C_1 = -\frac{1}{8}L^2$

$\therefore C_2 = \frac{5}{48}L^3$

$\therefore v = \frac{P}{EI}(\frac{1}{6}\langle x - \frac{L}{2}\rangle^3 - \frac{1}{8}L^2 x + \frac{5}{48}L^3)$

When $x = L$, $v(\uparrow) = -\frac{5}{48}\frac{P}{EI}$

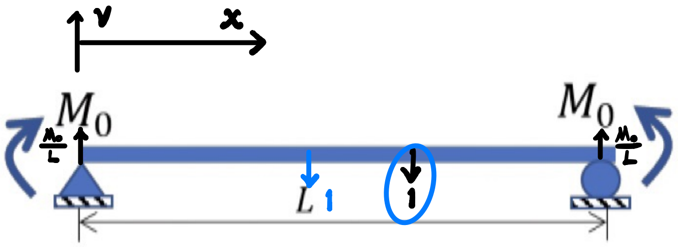
② $\bar{1} \cdot \delta = \int_0^L \frac{-P\langle x - \frac{L}{2}\rangle}{EI}(-x)\, dy$

$\therefore \delta = \frac{P}{EI}\int_0^{\frac{1}{2}L} 0\, dx + \frac{P}{EI}\int_{\frac{1}{2}L}^{L}(x^2 - \frac{L}{2}x)\, dx$

$= \frac{P}{EI}\int_{\frac{1}{2}L}^{L}(x^2 - \frac{1}{2}Lx)\, dx = \frac{P}{EI}[\frac{1}{3}x^3 - \frac{L}{4}x^2]_{\frac{1}{2}L}^{L}$

$= \frac{5}{48}\frac{P}{EI}$

2. Calculate the maximum deflection using both double integration and unit load approaches. Sketch the deflected shape.



b) $M = \frac{M_0}{L}x$

① $\frac{d^2v}{dx^2} = -\frac{1}{R} = -\frac{M}{EI} = -\frac{M_0}{EI}\cancel{\frac{x}{L}}$

$\therefore \frac{dv}{dx} = \int -\frac{M_0}{EI}\cancel{\frac{x}{L}}\,dx = -\frac{M_0}{EI}\left(\cancel{\frac{x^2}{2L} + C_1}\right)$ $x + C_1$

$\therefore v = -\frac{M_0}{EI}\int\left(\cancel{\frac{x^2}{2L} + C_1}\right)dx = -\frac{M_0}{EI}\left(\cancel{\frac{x^3}{6L} + C_1x + C_2}\right)$ $x + C_1$ $\frac{1}{2}x^2 + C_1x + C_2$

$\because x = 0$ and L, $v = 0$

$\therefore C_2 = 0$

$\therefore C_1 = \cancel{-\frac{1}{6}L}\ -\frac{1}{2}L$

$\therefore v = -\frac{M_0}{EI}\left(\cancel{\frac{x^3}{6L} - \frac{L}{6}x}\right)$ $\frac{1}{2}x^2 - \frac{1}{2}Lx$

$\frac{dv}{dx} = -\frac{M_0}{EI}\left(\cancel{\frac{x^2}{2L} - \frac{L}{6}}\right)$ $x - \frac{1}{2}L$

When $\frac{dv}{dx} = 0$, deflection goes maximum $(0 < x < L)$

$\therefore x = \cancel{\frac{\sqrt{3}}{3}L}\ \frac{1}{2}L$

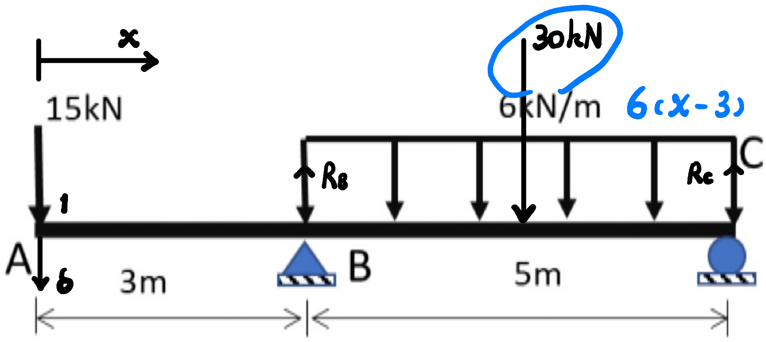
$\therefore v_{max} = \cancel{\frac{\sqrt{3}}{27}L^2\frac{M_0}{EI}}\ -\frac{1}{8}L^2\frac{M_0}{EI}$

② $\cancel{\bar{1}\cdot\delta = \int_0^L \frac{\frac{M_0}{L}x}{EI}\left(\frac{3-\sqrt{3}}{3}L - \langle x - \frac{\sqrt{3}}{3}L\rangle\right)dx}$

$\cancel{\therefore \delta = \frac{\sqrt{3}}{27}L^2\frac{M_0}{EI}}$

$\bar{1}\cdot\delta = \int_0^L \frac{M_0}{EI}\left(\frac{x}{2} - \langle x - \frac{L}{2}\rangle\right)dx = -\frac{1}{8}L^2\frac{M_0}{EI}$

3. Calculate the at deflection at A using the unit load approaches.



$R_B = 39kN$, $R_C = 6kN$

$M = -15x + 39\langle x-3\rangle - 30\langle x-5.5\rangle$

When $\bar{1}$ applied to A.

$R_B = \frac{8}{5}$

$\bar{M} = -x + \frac{8}{5}\langle x-3\rangle$

$\therefore\ \bar{1}\cdot\delta = \int_0^8 \frac{M}{EI}\bar{M}\,dx$

$$= \frac{1}{EI}\left(\int_0^3 (-15x - x)\,dx\ \cancel{+\int_3^{5.5}[-15x+39(x-3)][-x+\tfrac{8}{5}(x-3)]\,dx}\right.$$

$$6(x-3)\ \frac{x-3}{2}$$

$$+\int_{5.5\ 3}^{8}[-15x+39(x-3)-30(x-5.5)][-x+\tfrac{8}{5}(x-3)]\,dx$$

$$\cancel{= \frac{1}{EI}\left(\int_0^3 15x^2\,dx + \int_3^{5.5}\left(\frac{72}{5}x^2 - \frac{927}{5}x + \frac{2808}{5}\right)dx + \int_{5.5}^{8}\left(-\frac{18}{5}x^2 + \frac{288}{5}x - \frac{1152}{5}\right)dx\right)}$$

$$= \frac{1}{EI}\left(135 + \frac{525}{4}\ \cancel{\frac{825}{8} - \frac{75}{4}}\right)$$

$$= \cancel{-\frac{405}{8EI}}\ \frac{1065}{4EI}$$

$$\therefore\ \delta = \cancel{-\frac{405}{8EI}}\ \frac{1065}{4EI}$$